

MAL3040 — Statistics
Quiz 2 — with answers

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B.Sc.–B.Ed. III year

Time: 50 minutes

Maximum marks: 15 (+2 bonus)

The answers are printed in **this colour** below each part, briefly and with most of the working left out. If your own answer disagrees with one of them, go back to your script and find where the two of you parted company.

What you may use. Tables A1 to A4 from the appendix of Ross are supplied with this paper. No other formula is supplied.

Z is a standard normal random variable and Φ is its distribution function. The distribution χ_k^2 is that of $Z_1^2 + \dots + Z_k^2$ for independent standard normal Z_i ; the t distribution with k degrees of freedom is that of $Z/\sqrt{V/k}$ with $V \sim \chi_k^2$; and the F distribution with m and k degrees of freedom is that of $(V_1/m)/(V_2/k)$ with $V_1 \sim \chi_m^2$ and $V_2 \sim \chi_k^2$. In each case the numerator and the denominator are independent.

Question 1. Each part below asks for a single number. Use the tables.

- (a) Find $P(Z > -0.75)$. **[1]**

Ans. By symmetry, $P(Z > -0.75) = \Phi(0.75) = 0.7734$.

- (b) Let V have the χ^2 distribution with 8 degrees of freedom. Find a with $P(V < a) = 0.025$. **[1]**

Ans. $P(V > a) = 0.975$, so $a = \chi_{0.975, 8}^2 = 2.180$.

- (c) Let W have the F distribution with 3 degrees of freedom for the numerator and 5 for the denominator. Find the number c for which $P(W < c) = 0.05$. **[1]**

Ans. $c = F_{0.95, 3, 5} = 1/F_{0.05, 5, 3} = 1/9.01 = 0.1110$. The entry read has numerator 5 and denominator 3, the two exchanged.

[3 marks]

Question 2. A random variable X has the Bernoulli distribution with parameter p , that is, $P(X = 1) = p$ and $P(X = 0) = 1 - p$. In parts (b) and (c), $X_1, \dots, X_n$ is a random sample from this distribution.

- (a) Using the fact that $X^2 = X$, show that $\text{Var}(X) = p(1 - p)$. **[1]**

Ans. X takes only the values 0 and 1, and $0^2 = 0$, $1^2 = 1$, so $X^2 = X$ and hence $E[X^2] = E[X] = p$. Therefore $\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$.

- (b) Let $Y = X_1 + \dots + X_n$. Show that $\text{Var}(Y) = np(1 - p)$. Your working must say at which step independence is used. **[2]**

Ans. $\text{Var}(Y) = \text{Var}(X_1) + \dots + \text{Var}(X_n)$; this step is where independence is used, since it is independence that makes the covariance terms vanish. Each variance equals $p(1 - p)$ by part (a), so $\text{Var}(Y) = np(1 - p)$.

- (c) Show that $S^2 = \frac{n}{n-1} \bar{X}(1 - \bar{X})$. **[1]**

Ans. Each X_i is 0 or 1, so $X_i^2 = X_i$ and $\sum X_i^2 = \sum X_i = n\bar{X}$. Hence $\sum (X_i - \bar{X})^2 = \sum X_i^2 - n\bar{X}^2 = n\bar{X} - n\bar{X}^2 = n\bar{X}(1 - \bar{X})$, and dividing by $n - 1$ gives $S^2 = \frac{n}{n-1} \bar{X}(1 - \bar{X})$.

[4 marks]

Question 3. Two short derivations from the definitions.

- (a) Let T have the t distribution with k degrees of freedom. Show that T^2 has the F distribution with 1 and k degrees of freedom. [2]

Ans. Write $T = Z/\sqrt{V/k}$ with $Z \sim N(0, 1)$ and $V \sim \chi_k^2$ independent. Then $T^2 = \frac{Z^2}{V/k} = \frac{Z^2/1}{V/k}$. Now $Z^2 \sim \chi_1^2$, and it is independent of V . So T^2 is a χ_1^2 divided by its degrees of freedom, over an independent χ_k^2 divided by its degrees of freedom, which is the definition of the F distribution with 1 and k degrees of freedom.

- (b) Let T have the t distribution with 12 degrees of freedom. Using the symmetry of the t distribution and the table, find the number a for which $P(T < a) = 0.05$. [1]

Ans. Symmetry gives $t_{0.95, 12} = -t_{0.05, 12}$, and the table gives $t_{0.05, 12} = 1.782$, so $a = -1.782$.

[3 marks]

Question 4. Five observations have sample mean $\bar{x} = 10$. Four of the five deviations $x_i - \bar{x}$ are

$$2, \quad -3, \quad 1, \quad 4.$$

- (a) Write down the fifth deviation. [1]

Ans. The five deviations add up to 0, because $\sum(x_i - \bar{x}) = \sum x_i - n\bar{x} = 0$. The first four add up to 4, so the fifth deviation is -4 .

- (b) Suppose instead that you had been told four of the five deviations from the *population* mean μ , that is, four of the numbers $x_i - \mu$. Would these four determine the fifth? Give a *precise* reason. [1]

Ans. No. In part (a) the fifth deviation was determined because $\bar{x}$ is computed from the same five observations, and this forces $\sum(x_i - \bar{x}) = 0$. The population mean μ is a fixed number that is not computed from the data, so the five numbers $x_i - \mu$ carry no such constraint; their sum is $5(\bar{x} - \mu)$, which is not zero in general.

[2 marks]

Question 5. A trial is repeated 100 times, independently. Each trial gives a one with probability 0.02, and a zero otherwise. Let Y be the number of ones in the 100 trials. Round any standardised value to two decimal places before using the table.

- (a) Find $\text{Var}(Y)$. [1]

Ans. By Question 2(b) with $n = 100$ and $p = 0.02$, $\text{Var}(Y) = np(1 - p) = 100(0.02)(0.98) = 1.96$.

- (b) The central limit theorem approximates Y by a normal random variable with the same mean and the same variance as Y . Find the probability that this normal random variable is negative. [1]

Ans. The mean is $E[Y] = np = 2$, by the linearity of expectation, and the standard deviation is $\sqrt{1.96} = 1.4$. So the probability is $P(Z < (0 - 2)/1.4) = P(Z < -1.43) = 1 - 0.9236 = 0.0764$.

- (c) The number of trials is 100, well above the rule of thumb of 30 trials for applying the central limit theorem. Using your answer to part (b), say whether the central limit theorem may be applied here, and justify your conclusion. [1]

Ans. It may not. The number Y is a count, so it can never be negative, yet the approximating normal random variable takes negative values with probability 0.0764. A curve that puts nearly eight per cent of its probability where Y has none is misplaced, and its other probabilities cannot be trusted either. The number of trials by itself decides nothing; what has to be large is $np(1 - p) = 1.96$, or equivalently $np = 2$ and $n(1 - p) = 98$, and the usual working rule asks for about 10 or more.

[3 marks]

Bonus question. *Optional. These marks are over and above the fifteen marks of the paper.*

The definition of the t distribution requires that $Z \sim N(0, 1)$ and $V \sim \chi_k^2$ be independent. Give an example to show that if Z and V are not independent, then $Z/\sqrt{V/k}$ does not follow a t_k distribution. **[2]**

Ans. Take $k = 1$ and $V = Z^2$. By the definition of the χ^2 distribution, V has the χ_1^2 distribution, and it is plainly not independent of Z . Then

$$\frac{Z}{\sqrt{V/1}} = \frac{Z}{|Z|} = \begin{cases} +1 & \text{if } Z > 0, \\ -1 & \text{if } Z < 0, \end{cases}$$

each with probability $1/2$. A t random variable takes a whole range of values, so this is not one.

[2 bonus marks]